

Problem-Solution Fit

Date	26 June 2026
Team ID	
Project Name	Hydra Shield: An Intelligent Flood Prediction and Early Warning System (Rising Waters)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

1 CUSTOMER SEGMENT(S) [CS]  - Disaster Management Authorities - Government Agencies - Meteorological Departments - Emergency Response Teams - Residents in Flood-Prone Areas	6 CUSTOMER LIMITATIONS [CL]  - Limited access to real-time weather information - Incomplete environmental data - Difficulty predicting sudden weather changes - Lack of advanced forecasting tools - Delayed communication during emergencies	5 AVAILABLE SOLUTIONS PROS & CONS [AS]  Manual Monitoring: Simple but slow and less accurate. Traditional Rule-Based Systems: Fast but cannot adapt to changing weather conditions. Machine Learning Models: Accurate, scalable, adaptive and capable of learning from historical flood data.
2 PROBLEMS / PAINS + ITS FREQUENCY [PR]  - Delayed flood warnings - Inaccurate flood prediction methods - Loss of lives and property due to late alerts - Difficulty in monitoring changing weather conditions - Frequent flooding during heavy rainfall seasons	9 PROBLEM ROOT / CAUSE [RC]  - Extreme rainfall and climate change - Poor flood forecasting methods - Delayed data processing - Lack of timely warning systems - Inadequate disaster preparedness	7 BEHAVIOR + ITS INTENSITY [BE]  - Authorities require early and reliable flood predictions. - Citizens expect timely warning alerts. - Emergency teams need quick decision-making support. - Continuous monitoring is essential during heavy rainfall.
3 TRIGGERS TO ACT [TR]  - Heavy rainfall prediction - Rising river or water levels - Need for early evacuation planning - Increasing flood-related disasters - Demand for accurate real-time prediction	10 YOUR SOLUTION [SL]  Develop Hydra Shield: An Intelligent Flood Prediction and Early Warning System using Machine Learning algorithms (Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost) integrated with a Flask web application. The system analyzes weather and environmental parameters to predict flood risk (High Risk / Low Risk) in real time. By providing early warnings, it helps disaster management authorities and communities make informed decisions, improve preparedness, reduce response time, and minimize the impact of floods.	8 CHANNELS OF BEHAVIOR [CH]  ONLINE - Government Disaster Management Portals - Weather Monitoring Websites - Mobile Applications - Web-Based Flood Prediction System OFFLINE - Disaster Management Control Rooms - Emergency Response Centers - Public Warning Systems - Local Government Offices
4 EMOTIONS BEFORE / AFTER [EM]  Before: - Fear - Anxiety - Uncertainty - Property damage concerns - Lack of preparedness After: - Early awareness - Better preparedness - Increased safety - Faster emergency response - Greater public confidence		